

Marcos Hernandez

Computer Science & Business Information Systems

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Education

Eastern Connecticut State University - Willimantic, CT

B.S. Computer Science, Minor: **Data and Information Engineering**

B.S. Business Information Systems

May 2025

GPA 3.5

GPA 3.4

Relevant Courses

Business Intelligence & Data Solutions	Data Structures	AWS Data Engineering Foundations
Business Database Management (SQL)	Software Engineering	Data Mining & Machine Learning

Technical Skills

Programming & Scripting: Python (OOP), Java, SQL (PostgreSQL, MySQL), Bash/Zsh

Cloud & Data Platforms: AWS (Glue, S3, EC2, Lambda, DynamoDB), Snowflake

Data Analytics Tools: Tableau, IBM SPSS, Excel, Minitab, Weka

Development Tools: Jupyter Notebook, Git/GitHub, VS Code, Eclipse, Lucidchart, Jira

Professional Skills: Leadership, English/Spanish, Technical Communication, problem solving

Research & Projects

Senior Research: ETL Optimization (Spring 2025)

Evaluating Dynamic Mapping Matrix vs Traditional ETL Mapping for Optimized Data Processing

- Designed 3 **transformation pipeline** implementations on NYC Taxi data using **Python**.
- Simulated matrix based ETL logic via NumPy to emulate DMM sparsity and compaction.
- Achieved **>85% memory savings** and improved runtime with **benchmarking** controls.

Data Engineering Pipelines with Notebooks in Snowflake & AWS

- Built **scalable** ETL data pipelines with Python, **SQL**, Snowpark, and **AWS S3** external data.
- Automated workflows with GitHub Actions and Snowflake's Task DAG API, following **CI/CD**.
- Modeled data** using star schemas and monitored performance via Snowsight dashboards.

Software Classification Optimization Using Machine Learning

- Developed and evaluated 7 classification models (NB, MLP, kNN, SVM, LR, C4.5, RF100).
- Applied **feature engineering**, dimension reduction, and model evaluation in confusion matrices.
- Achieved **82.45% peak AUC** using 10-run, 5-fold cross-validation on telecom software datasets.
- Visualized** 28+ model results in **Tableau**, to **analyze** performance gains in feature reduction.

Experience

ECSU IT Department – Network Admin Intern | Willimantic, CT

Aug 2024 - Nov 2024

- Collaborated on a **\$7M** network infrastructure upgrade, completed one week ahead of schedule.
- Created a VoIP cube upgrade plan with **<30s downtime**, improving redundancy.
- Recovered and optimized **\$100K+** in hardware, cutting project costs and time.
- Balanced** large-scale fiber, switch, and software deployment projects with real-time IT support.

ECSU CSC Department – Teaching Assistant | Willimantic, CT

Jan 2024 - May 2024

Course: Introduction to Python and Machine Intelligence

- Supported 20+ students in Python, SDLC, **OOP**, and **data manipulation** using **Pandas**.
- Led hands on **data preprocessing** and ML lab sessions in Jupyter Notebook.
- Guided kNN model development using Scikit-learn, teaching evaluation via confusion matrices.

Leadership, Activities & Honors

- Tableau Ambassador** - Data Viz enthusiast group lead (20+ students) 2024-2025
- Upsilon Pi Epsilon** - National Honors Society in Computing Sciences 2024
- Mentor** - TheDream.US mentorship program 2022-2024
- TheDream.US Scholar** - \$80,000 scholarship recipient 2021-2025
- President & Treasurer** - ECSU-BIS Info. Tech. Students of America Chapter 2021-2023
- Valedictorian** - Savannah Classical Academy, Savannah GA 2021